DEVSPHERE

INTERNSHIP PROGRAM

WEEK 1 TASK REPORT

Artificial Intelligence (AI) & Machine Learning (ML)

Report: Week #1

Intern Name: Saud Masood

Domain: Artificial Intelligence & Machine Learning

Submitted To: DevSphere Team

Submission Date: 5 September 2026

1. Week 1 Task Brief

Program Requirement: All interns are required to complete their assigned Week 1 tasks according to their designation and present the work professionally with creativity and functionality.

Submission Requirements

- Prepare a PDF or Word document.
- Include the task question/instructions.
- Include the complete code.
- Include screenshots of code and program outputs.
- Include the dataset and explain the data-cleaning process.
- Clearly explain the completed work and results.
- Submit the completed work via email to devsphere1.info@gmail.com.

Deadline: 6 September 2026

2. Artificial Intelligence (AI) — Week 1

Task Question

Topic: AI Basics + Python
Task: Implement a Basic AI Logic Program
Requirement: Write a Python program using simple decision-making (if-else) logic. Example use cases include a recommendation system or basic chatbot logic.
Output: Python file with working logic.

Project Title

Student Career Recommendation System

Aim

To develop a basic rule-based AI system in Python that accepts user preferences and programming experience, applies predefined decision rules, and provides a suitable career recommendation.

Approach

The system follows the flow: User Input → Decision Rules → Recommendation → Output.

Python Code

```
print("======================================")
print("   AI Career Recommendation System")
print("======================================")

interest = input(
    "\nWhat are you interested in? "
    "(programming/design/business/data): "
).lower()

experience = input(
    "Do you have programming experience? (yes/no): "
).lower()

if interest == "programming" and experience == "yes":
    recommendation = "Software Developer"

elif interest == "programming" and experience == "no":
    recommendation = "Start learning Python and become a Software Developer"

elif interest == "data":
    recommendation = "Data Analyst / Machine Learning Engineer"

elif interest == "design":
    recommendation = "UI/UX Designer"

elif interest == "business":
    recommendation = "Business Analyst"

else:
    recommendation = "Explore different fields to discover your interest."

print("\n======================================")
print("          AI RECOMMENDATION")
```

```
print("======================================")

print("Your Interest:", interest)
print("Programming Experience:", experience)
print("Recommended Career:", recommendation)

print("\nAI Decision completed successfully!")
```

Code Screenshot — By Intern

```
#Ai code week 1
print("====================================")
print("   AI Career Recommendation System")
print("====================================")

interest = input(
    "\nWhat are you interested in? "
    "(programming/design/business/data): "
).lower()

experience = input(
    "Do you have programming experience? (yes/no): "
).lower()

if interest == "programming" and experience == "yes":
    recommendation = "Software Developer"

elif interest == "programming" and experience == "no":
    recommendation = "Start learning Python and become a Software Developer"

elif interest == "data":
    recommendation = "Data Analyst / Machine Learning Engineer"

elif interest == "design":
    recommendation = "UI/UX Designer"

elif interest == "business":
    recommendation = "Business Analyst"

else:
    recommendation = "Explore different fields to discover your interest."
```

```
elif interest == "programming" and experience == "no":
    recommendation = "Start learning Python and become a Software Developer"

elif interest == "data":
    recommendation = "Data Analyst / Machine Learning Engineer"

elif interest == "design":
    recommendation = "UI/UX Designer"

elif interest == "business":
    recommendation = "Business Analyst"

else:
    recommendation = "Explore different fields to discover your interest."

print("\n======================================")
print("          AI RECOMMENDATION")
print("======================================")

print("Your Interest:", interest)
print("Programming Experience:", experience)
print("Recommended Career:", recommendation)

print("\nAI Decision completed successfully!")
```

3. AI Output & Result

Output Screenshot — By Intern

```
======================================
   AI Career Recommendation System
======================================

What are you interested in? (programming/design/business/data): programming
Do you have programming experience? (yes/no): yes

======================================
          AI RECOMMENDATION
======================================
Your Interest: programming
Programming Experience: yes
Recommended Career: Software Developer

AI Decision completed successfully!
```

Result

The program demonstrates a basic AI decision-making system using conditional rules. It successfully maps user input to a predefined career recommendation.

4. Machine Learning (ML) — Week 1

Task Question

Topic: Python + Pandas
Task: Perform Data Analysis on a Dataset
Requirements: Load a dataset using Pandas; show the first 5 rows, dataset information and summary statistics; handle missing values; and perform basic data cleaning.
Output: Python script/notebook and cleaned dataset.

Project Title

Student Performance Data Analysis & Cleaning

Aim

To use Python and Pandas to load, inspect, analyze, clean, and export a student-performance dataset while demonstrating the fundamental data-preparation steps required before machine learning.

Dataset Overview

The dataset contains student names, ages, gender, and marks in Mathematics, English, and Science. Missing values are intentionally included in selected marks fields to demonstrate data cleaning.

Dataset — students.csv

Name	Age	Gender	Math	English	Science
Ali	20	Male	85.0	78.0	90
Ahmed	21	Male	72.0	80.0	75
Sara	20	Female	95.0	88.0	92
Ayesha	22	Female		91.0	89
Usman	21	Male	68.0		70
Hamza	20	Male	80.0	75.0	85
Fatima	21	Female	90.0	85.0	94
Bilal	22	Male	76.0	82.0	78
Hina	20	Female		89.0	91
Zain	21	Male	88.0	76.0	84

ML Python Code

```
import pandas as pd

print("=====================================")
print("       ML WEEK 1 - DATA ANALYSIS")
print("=====================================")

df = pd.read_csv("students.csv")

print("\nDataset loaded successfully!")

print("\n========== FIRST 5 ROWS ==========")
print(df.head())

print("\n========== DATASET INFO ==========")
df.info()

print("\n========== SUMMARY STATISTICS ==========")
print(df.describe())
```

```
print("\n========== MISSING VALUES ==========")
print(df.isnull().sum())

numeric_columns = df.select_dtypes(include="number").columns

for column in numeric_columns:
    df[column] = df[column].fillna(df[column].mean())

text_columns = df.select_dtypes(include="object").columns

for column in text_columns:
    df[column] = df[column].fillna("Unknown")

df = df.drop_duplicates()

print("\n========== CLEANED DATASET ==========")
print(df.head())

print("\n========== MISSING VALUES AFTER CLEANING ==========")
print(df.isnull().sum())

df.to_csv("cleaned_students.csv", index=False)

print("\nData cleaning completed successfully!")
print("Cleaned dataset saved as cleaned_students.csv")
```

5. ML Code Screenshot — By Intern

```python
#ml code
import pandas as pd

print("======================================")
print("       ML WEEK 1 - DATA ANALYSIS")
print("======================================")

df = pd.read_csv("students.csv")

print("\nDataset loaded successfully!")

print("\n========== FIRST 5 ROWS ==========")
print(df.head())

print("\n========== DATASET INFO ==========")
df.info()

print("\n========== SUMMARY STATISTICS ==========")
print(df.describe())

print("\n========== MISSING VALUES ==========")
print(df.isnull().sum())

numeric_columns = df.select_dtypes(include="number").columns
```

```python
numeric_columns = df.select_dtypes(include="number").columns

for column in numeric_columns:
    df[column] = df[column].fillna(df[column].mean())

    text_columns = df.select_dtypes(include="object").columns

    for column in text_columns:
        df[column] = df[column].fillna("Unknown")

        df = df.drop_duplicates()

        print("\n========== CLEANED DATASET ==========")
        print(df.head())

        print("\n========== MISSING VALUES AFTER CLEANING ==========")
        print(df.isnull().sum())

        df.to_csv("cleaned_students.csv", index=False)

        print("\nData cleaning completed successfully!")
        print("Cleaned dataset saved as cleaned_students.csv")
```

6. ML Output Screenshot — By Intern

```
=====================================
       ML WEEK 1 - DATA ANALYSIS
=====================================

Dataset loaded successfully!

========== FIRST 5 ROWS ==========
     Name  Age  Gender  Math  English  Science
0     Ali   20    Male  85.0     78.0       90
1   Ahmed   21    Male  72.0     80.0       75
2    Sara   20  Female  95.0     88.0       92
3  Ayesha   22  Female   NaN     91.0       89
4   Usman   21    Male  68.0      NaN       70

========== DATASET INFO ==========
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
```

```
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   Name     10 non-null     object
 1   Age      10 non-null     int64
 2   Gender   10 non-null     object
 3   Math     8 non-null      float64
 4   English  9 non-null      float64
 5   Science  10 non-null     int64
dtypes: float64(2), int64(2), object(2)
memory usage: 612.0+ bytes

========== SUMMARY STATISTICS ==========
             Age       Math    English    Science
count  10.000000   8.000000   9.000000  10.000000
mean   20.800000  81.750000  82.666667  84.800000
std     0.788811   9.361776   5.873670   8.038795
min    20.000000  68.000000  75.000000  70.000000
25%    20.000000  75.000000  78.000000  79.500000
50%    21.000000  82.500000  82.000000  87.000000
75%    21.000000  88.500000  88.000000  90.750000
max    22.000000  95.000000  91.000000  94.000000

========== MISSING VALUES ==========
```

```
•••  ========== MISSING VALUES ==========
Name       0
Age        0
Gender     0
Math       2
English    1
Science    0
dtype: int64

========== CLEANED DATASET ==========
     Name  Age  Gender  Math  English  Science
0     Ali   20    Male  85.0     78.0       90
1   Ahmed   21    Male  72.0     80.0       75
2    Sara   20  Female  95.0     88.0       92
3  Ayesha   22  Female   NaN     91.0       89
4   Usman   21    Male  68.0      NaN       70

========== MISSING VALUES AFTER CLEANING ==========
Name       0
Age        0
Gender     0
Math       2
English    1
Science    0
dtype: int64

Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv

========== CLEANED DATASET ==========
     Name  Age  Gender  Math  English  Science
0     Ali   20    Male  85.0     78.0       90
1   Ahmed   21    Male  72.0     80.0       75
2    Sara   20  Female  95.0     88.0       92
3  Ayesha   22  Female   NaN     91.0       89
4   Usman   21    Male  68.0      NaN       70
```

```
========== MISSING VALUES AFTER CLEANING ==========
Name       0
Age        0
Gender     0
Math       2
English    1
Science    0
dtype: int64

Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv

========== CLEANED DATASET ==========
     Name  Age  Gender   Math  English  Science
0     Ali   20    Male  85.00     78.0       90
1   Ahmed   21    Male  72.00     80.0       75
2    Sara   20  Female  95.00     88.0       92
3  Ayesha   22  Female  81.75     91.0       89
4   Usman   21    Male  68.00      NaN       70

========== MISSING VALUES AFTER CLEANING ==========
Name       0
Age        0
Gender     0
Math       0
English    1
Science    0
dtype: int64

Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv
```

```
Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv

========== CLEANED DATASET ==========
     Name  Age  Gender   Math  English  Science
0     Ali   20    Male  85.00     78.0       90
1   Ahmed   21    Male  72.00     80.0       75
2    Sara   20  Female  95.00     88.0       92
3  Ayesha   22  Female  81.75     91.0       89
4   Usman   21    Male  68.00      NaN       70

========== MISSING VALUES AFTER CLEANING ==========
Name       0
Age        0
Gender     0
Math       0
English    1
Science    0
dtype: int64

Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv

========== CLEANED DATASET ==========
     Name  Age  Gender   Math    English  Science
0     Ali   20    Male  85.00  78.000000       90
1   Ahmed   21    Male  72.00  80.000000       75
2    Sara   20  Female  95.00  88.000000       92
3  Ayesha   22  Female  81.75  91.000000       89
4   Usman   21    Male  68.00  82.666667       70
```

```
========== MISSING VALUES AFTER CLEANING ==========
Name       0
Age        0
Gender     0
Math       0
English    0
Science    0
dtype: int64

Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv

========== CLEANED DATASET ==========
     Name  Age  Gender   Math    English  Science
0     Ali   20    Male  85.00  78.000000       90
1   Ahmed   21    Male  72.00  80.000000       75
2    Sara   20  Female  95.00  88.000000       92
3  Ayesha   22  Female  81.75  91.000000       89
4   Usman   21    Male  68.00  82.666667       70

========== MISSING VALUES AFTER CLEANING ==========
Name       0
Age        0
Gender     0
Math       0
English    0
Science    0
dtype: int64

Data cleaning completed successfully!
Cleaned dataset saved as cleaned_students.csv
```

```
...
    ========== CLEANED DATASET ==========
         Name  Age  Gender   Math    English  Science
    0     Ali   20    Male  85.00  78.000000       90
    1   Ahmed   21    Male  72.00  80.000000       75
    2    Sara   20  Female  95.00  88.000000       92
    3  Ayesha   22  Female  81.75  91.000000       89
    4   Usman   21    Male  68.00  82.666667       70

    ========== MISSING VALUES AFTER CLEANING ==========
    Name       0
    Age        0
    Gender     0
    Math       0
    English    0
    Science    0
    dtype: int64

    Data cleaning completed successfully!
    Cleaned dataset saved as cleaned_students.csv

    ========== CLEANED DATASET ==========
         Name  Age  Gender   Math    English  Science
    0     Ali   20    Male  85.00  78.000000       90
    1   Ahmed   21    Male  72.00  80.000000       75
    2    Sara   20  Female  95.00  88.000000       92
    3  Ayesha   22  Female  81.75  91.000000       89
    4   Usman   21    Male  68.00  82.666667       70

    ========== MISSING VALUES AFTER CLEANING ==========
    Name       0
    Age        0
    Gender     0
    Math       0
    English    0
    Science    0
    dtype: int64

    Data cleaning completed successfully!
    Cleaned dataset saved as cleaned_students.csv
```

Required Analysis Covered

- Load dataset using Pandas.
- Display the first five rows using df.head().
- Display dataset information using df.info().
- Generate summary statistics using df.describe().
- Check missing values using df.isnull().sum().
- Fill missing numerical values using the column mean.
- Fill missing text values with 'Unknown'.
- Remove duplicate rows.
- Export the cleaned dataset as cleaned_students.csv.

7. Cleaned Dataset — cleaned_students.csv

The following is the cleaned dataset generated by the ML script. The original dataset contains missing values; the cleaning script fills those values and exports the result.

Name	Age	Gender	Math	English	Science
Ali	20	Male	85.0	78.0	90
Ahmed	21	Male	72.0	80.0	75
Sara	20	Female	95.0	88.0	92
Ayesha	22	Female	81.75	91.0	89
Usman	21	Male	68.0	82.66666666666667	70
Hamza	20	Male	80.0	75.0	85
Fatima	21	Female	90.0	85.0	94
Bilal	22	Male	76.0	82.0	78
Hina	20	Female	81.75	89.0	91
Zain	21	Male	88.0	76.0	84

CSV Screenshot — By Intern

8. Final Completion Summary

Requirement	Status	Implementation
AI basic decision-making	Completed	Python if/elif/else logic
AI recommendation system	Completed	Career recommendation
Python working file	Completed	ai_career_recommendation.py
Pandas dataset loading	Completed	pd.read_csv()
First 5 rows	Completed	df.head()
Dataset information	Completed	df.info()
Summary statistics	Completed	df.describe()
Missing-value analysis	Completed	df.isnull().sum()
Missing-value handling	Completed	Mean / Unknown
Basic data cleaning	Completed	Duplicate removal
Cleaned dataset	Completed	cleaned_students.csv

Files Prepared

- ai_career_recommendation.py
- ml_week1.py
- students.csv
- cleaned_students.csv
- This Word report

Conclusion

The Week 1 AI and ML assignments have been implemented according to the provided DevSphere requirements. The AI task demonstrates rule-based decision making, while the ML task demonstrates the essential Pandas workflow for loading, inspecting, analyzing, cleaning, and exporting a dataset. The intern should insert personal code, terminal/output, and CSV screenshots in the marked sections before final submission.

Prepared by Saud Masood

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